

Unit 7 Test 2 Review — Roots, Factors, Cubes, & Grouping

Name: _____

Date: _____

Period: _____

Sum: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Diff: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

SOAP: Same, Opposite, Always Positive

Perfect Cubes Reference:

$1^3 = 1 \quad 2^3 = 8 \quad 3^3 = 27 \quad 4^3 = 64 \quad 5^3 = 125$

$6^3 = 216 \quad 7^3 = 343 \quad 8^3 = 512 \quad 9^3 = 729 \quad 10^3 = 1000$

Part 1: Roots and Factors (A2.7.D)

1. Let $f(x) = x^3 - 2x^2 - 5x + 6$. Which statements are true? **Select TWO.**

A. $x - 1$ is a linear factor of f D. $x = -2$ is a zero of f B. $f(-1) = 0$ E. The remainder when $f(x) \div (x - 1)$ is zeroC. $(1, 0)$ is an x -intercept of f

2. Consider $x^3 + 2x^2 - 9x - 18$. Which are linear factors? Check one per row.

Expression	Is a Factor	Not a Factor
$(x + 2)$		
$(x - 3)$		
$(x + 3)$		
$(x - 9)$		

3. $p(x) = x^3 - 4x^2 - 7x + 10$ has a zero at $x = 1$.

a) Since $x = 1$ is a zero, the factor is _____.

b) Divide $p(x)$ by that factor and then factor completely.

Part 2: Factoring by Grouping (A2.7.D / A2.7.E)

4. $x^3 + 3x^2 + 4x + 12$

6. $4x^3 + 6x^2 - 10x - 15$

5. $2x^3 - 5x^2 - 18x + 45$

7. Factors of $x^3 - 6x^2 + 3x - 18$?

Select TWO.

A. $x + 6$

B. $x - 6$

C. $x^2 + 3$

D. $x^2 - 3$

E. $x - 3$

Part 3: Sum & Difference of Cubes (A2.7.E)

8. Cube Root Warm-up: Find the cube root of each term ($\sqrt[3]{term}$).

a) $\sqrt[3]{8} = \underline{\hspace{2cm}}$

b) $\sqrt[3]{125} = \underline{\hspace{2cm}}$

c) $\sqrt[3]{x^3} = \underline{\hspace{2cm}}$

d) $\sqrt[3]{27x^3} = \underline{\hspace{2cm}}$

e) $\sqrt[3]{64} = \underline{\hspace{2cm}}$

f) $\sqrt[3]{216} = \underline{\hspace{2cm}}$

9. Select TWO factors of $8x^3 - 27$: A. $2x - 3$ B. $2x + 3$ C. $4x^2 + 6x + 9$ D. $4x^2 - 6x + 9$ E. $4x^2 - 6x - 9$ **10.** Factors of $x^3 + 216$?Quadratic: $\underline{\hspace{2cm}}$

$x + 6$	$x - 6$
$x^2 + 6x + 36$	$x^2 - 6x + 36$
$x^2 + 36x + 6$	$x^2 - 6x - 36$

Linear: $\underline{\hspace{2cm}}$

11. Factor: $125x^3 - 8$

$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$

12. Factor: $27 + 64x^3$

$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$

Part 4: Mixed Practice — Choose Your Strategy!

13. $x^3 + 4x^2 - 9x - 36$

Strategy: $\underline{\hspace{3cm}}$

15. $3x^3 - 6x^2 + 5x - 10$

Strategy: $\underline{\hspace{3cm}}$

14. $343x^3 - 1$

Strategy: $\underline{\hspace{3cm}}$

16. True or False?

"If $f(2) = 0$, then $(x + 2)$ is a factor of $f(x)$."**Circle:** TRUE FALSEExplain: $\underline{\hspace{4cm}}$